Dynamic Probabilistic Risk Assessment of Cascading Outages

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***Abstract*—Cascading outages in transmission power systems leading to blackouts or major power disruption are due to complex mechanisms. Some of the key mechanisms are intrinsically linked to power system dynamics that static models cannot represent. Additionally, stochastic failures of the protection devices are another important factor. Therefore, a dynamic probabilistic risk assessment of cascading outages is necessary. However, most of the existing approaches are based on static power system models. This paper discusses the limitation of these static models, reviews the state of the art on dynamic probabilistic risk assessment of cascading outages and proposes several techniques to solve the current challenges. Probabilistic dynamic simulations are compared to deterministic dynamic simulations using two test systems. The results show that probabilistic simulations show various impacts on the cascading risks for those two test systems.**

***Index Terms*—Cascading outages, cascading failures, blackouts, dynamic power system simulation, probabilistic risk assessment.**

I. Introduction

Cascading outages in transmission power systems leading to blackouts or major system disturbances are complex phenomena. In particular, a large variety of mechanisms are involved in cascading outages: common mode failures and hidden failures that cause several initial outages, high static loads after power flow redistribution that could lead to additional thermal failures, no corrective actions or wrong operators actions due to lack of situational awareness, static currents or apparent impedance triggering relays, voltage/small-disturbance/transient/frequency instabilities, etc. [1].

An important factor contributing to the complexity of these phenomena is their stochasticity. For example, a fault on a line should lead to the outage of this line only if all protection systems act perfectly. Most of the time, it will be the case. But there is a non-negligible probability that a hidden failure in a protection device will lead to further outage. What happens after an initiating event is not deterministic but probabilistic. Several *scenarios* or sequences of events (e.g., additional outages) are possible and with different *probabilities*. Each scenario can lead to adverse *consequences*. The risk of a particular cascading outage scenario is defined by the product of its *likelihood* (probability or frequency) by its adverse consequences (loss of supplied power, energy not served) [2]. The global risk of cascading outages is the sum of the risks for all scenarios. It can be viewed as the mean adverse consequences due to cascading outages (i.e., mean loss of supplied power, mean energy not served).

The aim of a probabilistic risk assessment of cascading outages is not only the computation of the global risk, but also the computation of the probability and the consequences of each cascading scenario. It can reveal critical elements and critical cascading paths in the transmission power systems. Numerous methodologies have been developed [1], but they are in general based on a static model of power systems. They are thus unable to take into account the dynamic cascading phenomena. A few methodologies tackle these dynamic phenomena by using full dynamic simulations of the transmission power system (including protection systems), but still face several challenges [3]–[6].

The purposes of this paper are to emphasize the need of dynamic methodologies for cascading outage risk assessment and to study the importance of the consideration of stochastic phenomena in dynamic simulations. Therefore, section II discusses qualities and drawbacks of static methodologies, in perspective with typical cascading outage development. Section III reviews the state of the art on dynamic probabilistic risk assessment methodologies. Section IV presents main challenges of a convincing dynamic probabilist risk assessment methodology. Section V compares deterministic dynamic simulations and probabilistic dynamic simulations of cascading outages, using two test systems. Finally, Section VI presents conclusions from this study.

II. Motivation

A. Typical cascading outage development

There are in general two phases in a typical cascading failure leading to a blackout: slow cascade, ruled by the competition between thermal transients and operators, and fast cascade, ruled by electrical transients and protection systems. The blackout that occurred in Italy on September 28, 2003 [7] is a good example of this typical cascading outage. The initiating event was the tripping of a 380-kV line between Switzerland and Italy at 03:01. It increased the load on the neighboring lines. From 03:02 to 03:25, operators tried to eliminate the overloads. But the thermal transient was faster than operators. At 03:25:21, a second 380-kV line tripped (loaded then at 110%), due to a tree flashover. It ended the slow cascade and started the fast cascade: between 03:25:25

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and 03:25:28, three 220-kV lines were disconnected by over-current relays. The Italian grid then lost its synchronism with the UCTE main grid which entailed the disconnection of all remaining lines between Italy and UCTE by protection systems. The negative imbalance between generation and load in Italy caused a frequency drop. The primary control operation of the generating units increased the generation a few seconds after the event, but the frequency felt by 1.5 Hz in this time. All the pumping units in service were automatically disconnected and a large amount of load was automatically shed during the frequency transient. But approximately the same amount of generation was also lost during the transient for many reasons. Therefore, frequency felt below 47.5 Hz, resulting in the tripping of the generating units that were still in operation, leading to a blackout at 03:28:05.

B. Validity of static models

During the slow cascade, dynamic phenomena are not dominant. A static analysis is then valid and has a advantage to be much faster than a dynamic analysis. In the beginning of the fast cascade, disconnection of elements are due to high currents (over-current protections) or low apparent impedances (distance relays). A static model is still able to capture these phenomena. However, specific dynamic phenomena play a crucial role during the fast cascade and cannot be taken into account by a static model. Static models can therefore be proficient to model the beginning of cascading outages, but are not anymore valid when the dynamic cascading mechanisms become dominant. This is shown in [5], [8], [9]. The estimation of the loss of load due to a critical combinations of failures by static methodologies is thus not completely justified by the typical development of a cascading outage. This estimation can be nevertheless accurate, but it should be validated by dynamic models.

III. Literature Review

Two types of methodologies can be found in the literature: deterministic and probabilistic cascading outage models. During a cascading outage sequence, the first category considers that the power system will follow a deterministic evolution, while the second category considers that the power system will follow a probabilistic evolution.

A. Deterministic cascading outage models

To model both the power system dynamics and discrete protection actions is a challenging topic. Most of the current cascading outage models are in a deterministic manner, which assumes all system components are functioning ideally.

The authors in [10] described an initial approach using a system of differential-algebraic equations (DAE) with an additional set of discrete equations (a hybrid DAE system) to dynamically model cascading failures. Ref. [9] further developed a non-linear dynamic model that has the capabilities of modeling various discrete events such as protection relays (including delay feature) and islanding. In order to test the model, a batch of randomly selected $N-2$ contingencies for several different static load configurations were simulated. It found that the distribution of blackout sizes and event lengths from the simulator correlate well with historical trends. The authors also compared the new model with a quasi-steady state model (described in [11]) and found that the two models tend to agree for the early stages of cascading, but produce substantially different results for later stages. This proposed model allows one to test the impact of different load models and protective device assumptions, and suggests that load models can substantially impact cascading failure risks. Oak Ridge National Laboratory (ORNL) [12] is dedicated to seeking real-time protection analysis in order to prevent cascading failures by consolidating Siemens PSS/E and Computer Aided Protection Engineering (CAPE).

These mentioned work have not considered the stochastic behavior of cascading failures, for example the probabilistic tripping and hidden failures of protection systems.

B. Probabilistic cascading outage models

In order to consider the stochasticity of cascading outages that are induced by erroneous operation of protection systems, probabilistic risk assessment techniques are needed. The purpose of a probabilistic risk assessment is to compute the frequency and adverse consequences for each potential cascading scenario. Following a particular initiating event, the Event Tree (ET) analysis is a technique used to find potential cascading sequences and to compute their probabilities. The ET is a diagram that shows the initiating event and the failures or successes of the systems during the propagation of the cascade. Dynamic probabilistic analysis of cascading outages need to be based on Dynamic ETs (DETs), because cascading outages are driven by the continuous evolution of electrical variables (transitions are due to protection systems) [13], [14]. In DETs, disconnection of elements by their protection systems create branching points of the ET. DETs allows considering stochastic behavior of protection systems: failures, errors in measurements...

DETs were first introduced in [13], [14] for the analysis of cascading outages . The main aim was to identify corrective actions that could be taken after an initiating event in order to prevent cascading outages, and not the computation of consequences of each possible cascading scenario. Consequently, branchings in DETs are only due to corrective actions, and not to protection systems. Starting from the initiating event, the dynamic behavior of the system is simulated until a predefined set-point is crossed by the corresponding electrical variable. Two new branches are then generated: one corresponds to the implementation of the corrective action and the other one corresponds to the failure of this implementation. Dynamic behavior of the system is simulated for each branch, and the branching process is repeated each time a set-point is crossed. The probability to have a specific scenario (i.e. a specific branch in the tree) is given by the products of successive branching probabilities. The global risk is given by the sum of the risk of all scenarios (product of their probability by their consequences). As a few number of events are considered on

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definite set-points, the number of possible scenarios is limited. This methodology is applied to a small test system with 6 generators, 21 buses and 31 branches.

The concept of DETs was then used to perform probabilistic risk assessments of cascading outages in [15]–[17]. The aim of these works was to include the stochastic behavior of protection systems. In [15], skeleton-based Monte Carlo Discrete DETs (DDETs) are used to simulate the stochastic behavior of a power system after different initiating events. Possible failures of protection systems, as well as error measurements and set-points misconfiguration are considered. In [16], dynamic simulations are used in the second level of a two-level probabilistic risk assessment of cascading outages. Only possible failures of protection systems on precise set-points are considered and the number of possible branches is therefore limited. Both [15], [16] are based on dynamic simulations implemented in Matlab, which limits the computation speed. On the contrary, [17] makes use of a commercial simulator, Eurostag [18], which improves the computation speed. A Monte Carlo simulation method is used. Failures of protection systems and their set-points are sampled at the beginning of each run according to their probability laws (Fig. 1). The dynamic behavior of the system is then simulated, including the action of protection systems, until the system reaches an electrical stable state or a blackout state. For each run, the simulation is hence purely deterministic. All these works are based on variants of the Kundur's two-area test system.

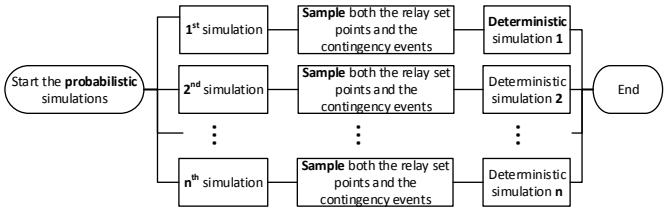


Fig. 1. The Monte Carlo simulation framework with probabilistic configuration.

C. Discussion

These two type of methods are complementary and show the limitations of the current state-of-the-art. It is common to analyze cascading outages deterministically in large power systems, and there are also works that study cascading outages probabilistically in small size power systems. However, no methodology currently exists to investigate dynamic cascading outages probabilistically in large power systems.

IV. Challenges and Perspectives

A. Computation time

One of the difficulties to apply dynamic probabilistic risk assessment methodologies to real power systems is the huge amount of computation time needed, even when efficient dynamic simulators are used. A double explosion of scenarios to analyze entails this difficulty: the large number of sets of initial contingencies to analyze and the large number of possible sequences for each set of contingencies . As modern power grids are operated according to the $N-1$ security rule, it is unlikely that a single contingency triggers a fast collapse of the grid. Consequently, the impact of $N-2$ contingencies must be analyzed. Real transmission systems have several thousands of elements. The number of $N-2$ contingencies is thus several millions. For each set of initial contingencies, there is only one "deterministic evolution" (when all protection systems work perfectly), but there is a huge number of possible probabilistic evolutions. The typical computation time needed to simulate one particular evolution of the fast cascade on a real power system is several tens of seconds. It is therefore unfeasible to simulate millions of different evolutions. Different numerical techniques have been proposed in the nuclear sector to limit the number of scenarios to analyze [19]: restriction of the possible branching points in time (i.e., discretization of time) with DDETs, Monte Carlo simulation (technique used in [17]) or combination of both with either skeleton-based Monte Carlo DDET (technique used in [15]) or Monte Carlo DET (MCDET) (technique described in [20] for the nuclear sector). The efficiency of these techniques for the dynamic probabilistic risk assessment of cascading outages should be evaluated.

B. Numerical convergence

Dynamic simulations of transmission power systems are based on the numerical resolution of DAEs. At some points, the iterative method used to solve the DAEs can fail to converge. There are mainly two reasons: no solution exists or the algorithm fails to find the existing solution. Commercial software implements methods to avoid the second reason (i.e. failure of the algorithm to find a solution). On the contrary, the first one (i.e. no solution exists) occurs in the context of cascading outages because the behavior of the system is simulated under extreme conditions. The non-linearity of algebraic equations is often due to loads. The existence of a solution is hence mainly related to the load model used. If the dynamic simulation reaches a point where no solution exists anymore, the usual assumption is to consider that the system endures a total collapse. However, in that case, the model is used outside its validity domain. The assumption of a total collapse can lead to an overestimation of the loss of supplied power. For example, if the absence of solution is due to a specific bus weakly connected to the system, only this bus will endure a voltage collapse, and not the overall system. Consequently, a dynamic simulation of cascading outages requires load models able to represent realistically load behaviors not only under normal conditions, but also under extreme conditions.

C. Reliability of protection systems

Reliability of protection systems in the context of cascading outages was first studied in [21], with static simulations in mind. Protection misoperations can be classified in two categories: missing trips (a protection system does not trip and it should) and unwanted trips (a protection system trips and it

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should not) [22]. For static simulations of cascading outages, the precise timing of events is not important. Therefore, it is sufficient to know the probability to have a missing trip and the probability to have an unwanted trip. However, for dynamic simulations, it is not sufficient to know that a protection system will disconnect an element, but we must know when, as the precise timing can be crucial . Consequently, the possible reasons of unwanted trips must be modeled and quantified. No detailed data about these stochastic phenomena are currently available. More research effort should be put on this challenge.

V. Case Studies

Ref. [17] showed that, for a particular initiating event and a particular load level, the consideration of erroneous operation of protection systems has a significant impact on the risk. However, it is not clear if it is important to consider these erroneous operations when the aim is the evaluation of the global risk due to numerous sets of initiating events. This section studies this point by performing a comparison between dynamic deterministic simulations and dynamic probabilistic simulations on two test systems. The Monte Carlo method described in section III-B is used for probabilistic simulations, similarly to [17].

A. Two-area test system

The first case study is based of the adaptation of the *Kundur's Two-Area System* [23] given in [17]. There are 8 generators with a maximal power of 400 MW for each of them. The peak load is 2758 MW. Overhead lines are protected with overcurrent and distance relays. There are under-frequency and under-voltage load shedding as system protection scheme. Power plants are protected with over-excitation, under-voltage, under-frequency, over-frequency and loss-of-synchronism relays.

Fig. 2 shows the Complementary Cumulative Distribution Functions (CCDF) of the loss of load when the impact of $N-2$ line contingencies is simulated for a random load between 50% and 100% of the peak load. The dynamic deterministic simulation assumes that protection systems act as expected (no stochasticity in the course of the transient). In the simplified probabilistic simulation, we take into account only the variability of set points and measurement errors by assuming that set points are considered distributed normally around their nominal value with a standard deviation equal to 1% of it. In the complete dynamic probabilistic simulation, we take also into account the possible failures of circuit breakers (the probability of failure of lines circuit breakers is 10^{-3} and the probability of failure of load shedding circuit breakers is 10^{-2}) and possible timer failures that can entail unwanted immediate trips and unwanted delayed trips (the probabilities of each of this possible erroneous operation are 2.5×10^{-3} for line protection systems, and 5×10^{-3} for load shedding relays and for generator protection systems). 5000 Monte Carlo runs are performed for each type of simulation. The Eurostag software is used to simulate the dynamic evolution (including the action of protection) of the system for each run. Erroneous operations of protection systems increase significantly the probability to have small loss of loads, but does not seem to have a significant impact on the probability of large blackouts.

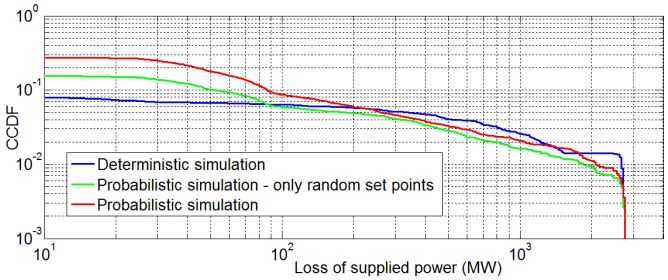


Fig. 2. Complementary cumulative distribution function of the loss of load for the two-area test system.

B. 2383-bus test system

The 2383-bus test system is the representative system of Polish network for the winter 1999-2000 [24]. The system includes 2383 buses, 2896 branches, 327 generators and 1826 loads, and it is a reasonable candidate for cascading outage simulation. This system is studied in the non-linear dynamic model developed in [9]. Temperature relay — monitoring the line temperatures that are computed by a differential equation — as well as distance relays protect the the overhead lines. Under-voltage and under-frequency load shedding relays are incorporated to release system stress if necessary.

The dynamic deterministic simulation assumes that relay (e.g., load shedding and distance relays) thresholds are exactly nominal value and that relays act as expected. In the probabilistic simulation, settings errors and measurements errors are considered by assuming that relays settings are normally distributed around their nominal value with a standard deviation equal to 1% of the nominal value. Similarly to the previous case study, in the probabilistic simulation, the probability of lines protection failure is 10^{-3} and the probability of load shedding failure 10^{-2}. Loads are assumed to be constant power loads. The impact of 1200 randomly selected $N-2$ line contingencies is dynamically simulated, both in a deterministic way and in a probabilistic way.

Fig. 3 depicts the CCDF of the loss of load for this comparison. The stochastic settings and failures of protection system tend to increase the probabilities of both small and large sizes of loss of load, as can be seen in the two zoomed subplots. In particular, it has more impacts on the small sizes (≤ 10 MW). The average loss of load for these 1200 simulations increases from 848.21 MW to 868.61 MW. However, the estimations of the probabilities to have a loss of supplied power between 10 MW and 100 MW are nearly equivalent for the deterministic simulation and the probabilistic one. Fig. 4 shows the CCDF of the total event length. We can see that the simulations with deterministic settings are more likely to have longer events than the ones with probabilistic settings, except the both ends of traces.

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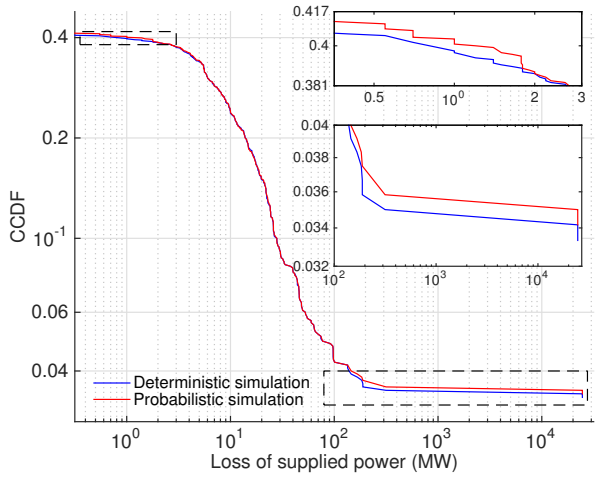


Fig. 3. Complementary cumulative distribution function of the loss of load for the 2383-bus test system.

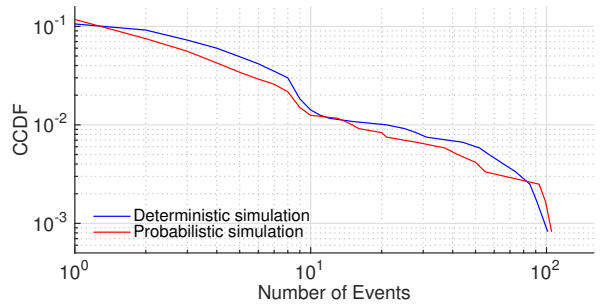


Fig. 4. Complementary cumulative distribution function of the event length for the 2383-bus test system.

VI. Conclusions

Among all cascading outage mechanisms, dynamic phenomena play a crucial role, especially during the second stage of typical cascading outages. Only a few methodologies have been developed to consider these dynamic phenomena by performing dynamic simulations. No methodology able to perform a dynamic probabilistic risk assessment considering the stochasticity of protection systems on real transmission power systems currently exists. However, the need to consider stochasticity of protection systems in dynamic simulations depends on the aim of the risk assessment. If the goal is to estimate global risk, this stochasticity does not seem to have a significant impact in large power systems. But, if the risk associated to a particular set of initiating contingencies must be estimated, it can be crucial to include this stochastic behavior in the dynamic simulation. Efficient techniques must therefore be developed to be able to perform dynamic probabilistic risk assessments on real power systems.

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